

Here is your **Ultimate 50-Question Master Mock Paper**. Mastering this single paper will ensure you are prepared for almost every major pattern the board throws at you.

CHEMISTRY (THEORY) — ULTIMATE 50-QUESTION MASTER PAPER

Maximum Marks: 120 (Extended format for practice) | Time: 4.5 Hours

SECTION A: Multiple Choice & Assertion-Reason (1 Mark Each)

These questions cover the most repeated foundational concepts.

- Solutions:** Which of the following solutions will have the highest boiling point in water? (Assuming complete dissociation)
(a) 1% KCl (b) 1% Glucose (c) 1% Urea (d) 1% CaCl_2
- d-Block:** Which of the following transition metal ions is not coloured?
(a) Cu^+ (b) Ni^{2+} (c) Co^{2+} (d) V^{3+}
- Electrochemistry:** Which of the following cells was used in the Apollo space programme?
(a) Mercury cell (b) Daniel cell (c) $\text{H}_2\text{-O}_2$ Fuel cell (d) Dry cell
- Kinetics:** The activation energy (E_a) of a reaction can be determined from the slope of which of the following plots?
(a) $\ln k$ vs. T (b) $\ln(k/T)$ vs. T (c) $\ln k$ vs. $1/T$ (d) $T/\ln k$ vs. $1/T$

5. **Solutions:** 1 mole of liquid A and 2 moles of liquid B make a solution having a total vapour pressure of 40 torr. The vapour pressures of pure A and pure B are 45 torr and 30 torr respectively. The above solution:

(a) is an ideal solution. (b) shows positive deviation. (c) shows negative deviation. (d) is a maximum boiling azeotrope.

6. **Amines:** Which of the following would not be a good choice for reducing nitrobenzene to aniline?

(a) LiAlH_4 (b) H_2/Ni (c) Fe and HCl (d) Sn and HCl

7. **Haloalkanes:** Which of the following haloalkanes react with aqueous KOH most rapidly by $\text{S}_{\text{N}}1$ reaction?

(a) 2-Chlorobutane (b) 1-Bromobutane (c) 2-Bromo-2-Methylpropane (d) 2,2-Dimethyl-1-Chloropropane

8. **Coordination:** The magnetic moment of $[\text{NiCl}_4]^{2-}$ is (Atomic number: Ni = 28):

(a) 1.82 BM (b) 2.82 BM (c) 4.42 BM (d) 5.46 BM

9. **Biomolecules:** The specific sequence in which amino acids are arranged in a protein is called its:

(a) Primary structure (b) Secondary structure (c) Tertiary structure (d) Quaternary structure

10. **Kinetics:** If the rate of a reaction increases sixteen times when the concentration of the reactant increases four times, the order of the reaction is:

(a) 2.5 (b) 2.0 (c) 1.5 (d) 0.5

11. **Amines:** Which of the following has the lowest pK_{a} value?

(a) $\text{CH}_3\text{-COOH}$ (b) $\text{O}_2\text{N-CH}_2\text{-COOH}$ (c) $\text{Cl-CH}_2\text{-COOH}$ (d) HCOOH

12. **Biomolecules:** Hydrolysis of sucrose is called:

(a) Inversion (b) Hydration (c) Esterification (d) Saponification

13. **Alcohols:** The reaction $\text{R-OH} + \text{Na} \rightarrow \text{R-ONa}^+ + \frac{1}{2}\text{H}_2(\text{g})$ suggests that alcohols are:

(a) Acidic (b) Basic (c) Neutral (d) Amphoteric

14. **d-Block:** Which property of transition metals enables them to behave as catalysts?

(a) High melting point (b) High ionisation enthalpy (c) Alloy formation (d) Variable oxidation states

15. **Haloalkanes:** Identify the chiral molecule among the following:

(a) 1-Bromobutane (b) 2-Bromobutane (c) 2-Bromo-2-methylpropane (d) 1-Bromo-3-methylbutane

(For Q16-20: Read the Assertion (A) and Reason (R). Choose correct option: A (Both true, R explains A), B (Both true, R does not explain A), C (A true, R false), D (A false, R true))

16. **Assertion (A):** The units of rate constant of a zero order reaction and rate of reaction are the same. **Reason (R):** In zero order reaction, the rate of reaction is independent of the concentration of reactants.

17. **Assertion (A):** Inversion of configuration is observed in S_N2 reaction. **Reason (R):** The reaction proceeds with the formation of carbocation.

18. **Assertion (A):** A mixture of o-nitrophenol and p-nitrophenol can be separated by steam distillation. **Reason (R):** o-nitrophenol is steam volatile due to intermolecular hydrogen bonding.

19. **Assertion (A):** Zr and Hf are of almost similar atomic radii. **Reason (R):** This is due to Lanthanoid contraction.

20. **Assertion (A):** Proteins are polymers of α -amino acids connected by a peptide bond. **Reason (R):** A tetrapeptide contains 4 amino acids linked by 4 peptide bonds.

SECTION B: Very Short Answer (2 Marks Each)

These are direct reasoning and conceptual questions.

21. **Electrochemistry:** State Kohlrausch's law of independent migration of ions. How does the molar conductivity of a strong electrolyte vary with dilution?

22. **Organic Conversions:** Give the chemical equation for Sandmeyer's reaction.

23. **Acidity:** Why is phenol more acidic than cyclohexanol?

24. **Biomolecules:** Give the reaction of glucose with Hydrogen Cyanide (HCN). Presence of which group is confirmed by this reaction?

25. **Coordination:** Write the IUPAC name of the linkage isomer of the complex $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$.

26. **Haloarenes:** Why are haloarenes less reactive towards nucleophilic substitution reactions than haloalkanes?

27. **Biomolecules:** The two strands in DNA are not identical but complementary. Explain.

28. **Kinetics:** State a condition under which a bimolecular reaction is kinetically a first-order reaction. Give an example.

29. **Electrochemistry:** What should be the signs (positive/negative) for E°_{Cell} and ΔG° for a spontaneous redox reaction occurring under standard conditions?

30. **Solutions:** What is meant by positive deviation from Raoult's law? What type of azeotrope is formed by positive deviation?

SECTION C: Short Answer (3 Marks Each)

These require multi-step numerical calculations or mechanisms.

31. **Electrochemistry Numerical:** Calculate the emf of the following cell at 298 K: $\text{Mg(s)} \mid \text{Mg}^{2+}(0.001 \text{ M}) \parallel \text{Cu}^{2+}(0.0001 \text{ M}) \mid \text{Cu(s)}$. Given $E^\circ_{\text{Mg}^{2+}/\text{Mg}} = -2.37 \text{ V}$, $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +0.34 \text{ V}$.

32. **Kinetics Proof:** Show that in the case of a first-order reaction, the time taken for completion of 99% reaction is twice the time required for 90% completion of the reaction. ($\log 10 = 1$)

33. **Solutions Numerical:** 0.3 g of acetic acid (CH_3COOH) dissolved in 30 g of benzene shows a depression in freezing point equal to 0.45°C . Calculate the percentage association of acid if it forms a dimer in the solution. (K_f for benzene = $5.12 \text{ K kg mol}^{-1}$).

34. **Kinetics Numerical:** The rate constant of a reaction quadruples when the temperature changes from 300 K to 320 K. Calculate the activation energy (E_a) for this reaction.

35. **Mechanism:** Give the mechanism of the acid-catalyzed dehydration of ethanol to yield ethene at 443K. (Or hydration of ethene to yield ethanol).

36. **Coordination (VBT):** $[\text{Ni}(\text{CO})_4]$ has tetrahedral geometry while $[\text{Ni}(\text{CN})_4]^{2-}$ has square planar geometry, yet both exhibit diamagnetism. Explain using Valence Bond Theory.

37. **Distinction Tests:** Give a chemical test to distinguish between: (i) Phenol and Ethanol (ii) Ethanal and Ethanoic acid (iii) Propan-2-one and Pentan-3-one.

38. **d-Block Trends:** Account for the following:

- (i) The $E^\circ (\text{Mn}^{3+}/\text{Mn}^{2+})$ value for manganese is highly positive.
- (ii) Actinoids show a wide range of oxidation states.
- (iii) Transition metals have high melting points.

39. **Organic Puzzle:** A compound (A) with molecular formula $\text{C}_4\text{H}_9\text{I}$ (primary alkyl halide), reacts with alcoholic KOH to give compound (B). Compound (B) reacts with HI to give (C) which is an isomer of (A). Write the structures of (A), (B), and (C).

40. **Electrochemistry:** The resistance of a conductivity cell filled with 0.2 M KCl solution is 200 Ω . If the resistance of the same cell when filled with 0.05 M KCl solution is 620 Ω , calculate the conductivity and molar conductivity of 0.05 M KCl solution. (Conductivity of 0.2 M KCl is 0.0248 S cm⁻¹).

SECTION D: Case-Based Questions (4 Marks Each)

Read the concepts carefully and apply them.

41. Case Study: Crystal Field Theory (CFT) and Colour

The Crystal Field Theory (CFT) of coordination compounds is based on the effect of different crystal fields on the degeneracy of d-orbital energies. The splitting of the d-orbitals provides different electronic arrangements.

- (a) On the basis of CFT, explain why the $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ complex is coloured. What happens on heating it? (2)
- (b) What is crystal field splitting energy? (1)
- (c) Why are low spin tetrahedral complexes rarely observed? (1)

42. Case Study: Nucleophilic Substitution ($\text{S}_{\text{N}}1$ vs $\text{S}_{\text{N}}2$)

In $\text{S}_{\text{N}}1$ reactions, the polarity of the system increases from reactant to transition state, meaning polar solvents accelerate the reaction. Weak polar solvents are favorable for $\text{S}_{\text{N}}2$.

- (a) Why does racemisation occur in $\text{S}_{\text{N}}1$? (1)
- (b) Why is ethanol less polar than water? (1)
- (c) Arrange the following in the increasing order of their reactivity towards $\text{S}_{\text{N}}1$ reactions: 2-Bromo-2-methylbutane, 1-Bromopentane, 2-Bromopentane. (2)

43. Case Study: Conductivity and Batteries

Rahul set up an experiment to find the resistance of aqueous KCl solution at 298 K using a Wheatstone bridge.

- (a) Why does conductivity (κ) decrease with dilution? (1)

- (b) If Λ_m° of KCl is $150.0 \text{ S cm}^2 \text{ mol}^{-1}$ and Λ_m is 141.0, calculate the degree of dissociation (α) of 0.01 M KCl. (1)
- (c) Write the reactions taking place at the anode and cathode of a Lead Storage Battery (Inverter battery) when it is in use. (2)

SECTION E: Long Answer Questions (5 Marks Each)

These are high-scoring, multi-part questions.

44. d- and f-Block Elements:

- (a) Explain the reason why Tungsten (W) and Chromium (Cr) have the highest melting points in their series, and why Manganese (Mn) has a sharp dip in its melting point. (2)
- (b) Complete the following ionic equation and justify that it is a disproportionation reaction: $3\text{MnO}_4^{2-} + 4\text{H}^+ \rightarrow$ _____ + _____ + $2\text{H}_2\text{O}$. (2)
- (c) How do you prepare potassium manganate from pyrolusite ore? (1)

45. Aldehydes, Ketones, and Carboxylic Acids:

An organic compound (X) having molecular formula $\text{C}_5\text{H}_{10}\text{O}$ can show various properties depending on its structures.

- (i) Draw the structure of (X) if it shows the Cannizzaro reaction. (1)
- (ii) Draw the structure of (X) if it reduces Tollens' reagent and has a chiral carbon. (1)
- (iii) Draw the structure of (X) if it gives a positive iodoform test. (1)
- (iv) Write the chemical equations for the Clemmensen reduction and Etard reaction. (2)

46. Electrochemistry:

- (a) Calculate the emf of the following cell at 25°C :



(Given: $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}$). (3) (Watch out: H^+ is squared in Q!)

- (b) State Faraday's second law of electrolysis. (1)
- (c) How much electricity is required in terms of Faraday for the reduction of 1 mol of $\text{Cr}_2\text{O}_7^{2-}$ to Cr^{3+} ? (1)

47. Solutions:

- (a) Why is the boiling point of a 1M NaCl solution more than that of a 1M glucose solution? (1)
- (b) A non-volatile solute 'X' (molar mass = 50 g/mol) when dissolved in 78g of benzene reduced its vapour pressure to 90%. Calculate the mass of X dissolved in the solution. (2)
- (c) Calculate the boiling point elevation for a solution prepared by adding 10g of MgCl_2 to 200g of water, assuming MgCl_2 is completely dissociated. (K_b for Water = $0.512 \text{ K kg mol}^{-1}$). (2)

48. Chemical Kinetics:

The following experimental rate data were obtained for a reaction: $\text{A(g)} + \text{B(g)} \rightarrow \text{C(g)} + \text{D(g)}$

EXP	[A]	[B]	RATE
EXP 1	3.0×10^{-2}	2.0×10^{-2}	1.89×10^{-4}
EXP 2	3.0×10^{-2}	4.0×10^{-2}	1.89×10^{-4}
EXP 3	6.0×10^{-2}	4.0×10^{-2}	7.56×10^{-4}

- (i) Determine the order of reaction with respect to A and B. (2)
- (ii) Write the rate law equation. (1)
- (iii) Calculate the value of the rate constant (k) and write its units. (2)

49. Amines:

- (a) Arrange the following in increasing order of their basic strength in aqueous solution: $\text{CH}_3\text{-NH}_2$, $(\text{CH}_3)_2\text{NH}$, $(\text{CH}_3)_3\text{N}$. (1)
- (b) Why is the $\text{p}K_b$ value of aniline more than that of methylamine? (1)
- (c) How can you convert the following? (2)

- (i) Ethanoic acid to methanamine.
- (ii) Propanenitrile to 1-aminopropane.
- (d) Give the structures of A and B in the following reaction:



50. Organic Reasoning & Properties:

- (a) Why are the α -hydrogens of aldehydes and ketones acidic in nature? (1)
- (b) An organic compound 'A' with molecular formula $\text{C}_4\text{H}_8\text{O}_2$ undergoes acid hydrolysis to form two compounds 'B' and 'C'. Oxidation of 'C' with acidified potassium permanganate also produces 'B'. Sodium salt of 'B' on heating with soda lime gives methane.
 - (i) Identify 'A', 'B', and 'C'. (3)
 - (ii) Out of 'B' and 'C', which will have a higher boiling point? Give reason. (1)